

which consists of unrestricted and restricted endowment, represents 46% of total endowment. *Term endowment*, which is about the same size as unrestricted endowment, represents nonperpetual funds. *Quasi-endowment* constitutes 23% of total endowment and does not come from gifts, but from other sources of university income like operating surpluses. Quasi-endowment is unrestricted, but it cannot be spent until a board reclassifies it as true endowment.

The most interesting part of total endowment is the “other” category. This is a large category representing \$94.2 billion, or almost one quarter, of total endowment wealth. It includes artwork, farmland, patents, trademarks, and royalties. The last three result from commercial application of research projects and are now large sources of revenue for universities. The “Axel patents,” for example, generated close to \$800 million in revenue for my home institution, Columbia University, and the inventors. The Axel patents, now expired, showed how foreign DNA could be inserted into a cell to produce certain proteins. They are named after Richard Axel, a professor of neuroscience who won the Nobel Prize in Physiology or Medicine in 2004 for work (quite unrelated to the Axel patents) on how the brain interprets smell.

4.2. UNIVERSITIES ARE INVESTING PIONEERS

The leading university endowments have long been investing pioneers. During the 1920s and 1930s, despite the Great Depression, universities shifted toward equities.⁵⁸ We remember John Maynard Keynes as one of the greatest economists of the twentieth century, but lesser known is that he was also an endowment manager. Keynes managed the endowment of King’s College, Cambridge, and was one of the pioneers of the trend to move into equities.⁵⁹ These greater equity allocations proved hugely beneficial in the long run, but at the time many questioned the sanity of endowment managers. It turned out that the Great Depression was the perfect time to buy equities, as prices were low and expected returns were high—but buying when everyone was selling took guts.

During the 1980s the leading universities—Harvard, Princeton, and Yale (HPY)—started to shift into alternative assets, especially more illiquid asset classes. Yale, led by David Swensen, moved first: by reducing its public equity allocation from 60% in 1980 to 20% by the mid-1990s. Its allocation to private equity, meanwhile, shot up from below 5% to above 20%.⁶⁰ Swensen’s (2009) thesis, called the *endowment model*, was that long-lived investors had an edge in

⁵⁸ See Goetzmann, Griswold, and Tseng (2010).

⁵⁹ As described by Chambers and Dimson (2012).

⁶⁰ Numbers from Goetzmann and Oster (2012, Figures 2 and 3). See also Lerner, Schoar, and Wong (2008).